

UMM AL MARADIM, Kuwait

WMO: 405960

Lat: 28.680N

Lon: 48.652E

Elev: 27

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.6	54.1	29.8	23.9	55.1	33.4	28.0	57.3	56.3	64.4	49.9	60.7	21.4	310	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.4	103.6	81.3	101.1	80.6	99.4	80.7	89.7	94.9	88.7	94.5	87.8	94.2	10.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
88.4	206.2	93.7	87.3	199.0	93.6	86.2	191.9	93.5	54.8	95.2	53.5	94.8	52.3	94.2	93.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
44.1	36.4	30.4	DB	48.3	108.1	2.7	3.3	46.3	110.5	44.7	112.5	43.1	114.3	41.1	116.7	
			WB	41.3	90.4	2.7	1.5	39.4	91.5	37.8	92.4	36.2	93.2	34.3	94.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.8	61.4	62.6	67.9	74.8	84.7	92.9	95.2	95.9	93.7	87.0	74.9	65.9
	DBStd	13.22	4.28	3.52	3.28	3.23	4.13	3.18	2.39	2.20	2.11	3.22	4.81	4.05
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	250	121	78	10	0	0	0	0	0	0	0	2	39
	CDD50	10888	353	353	555	743	1077	1287	1403	1422	1311	1146	746	492
	CDD65	5663	9	11	100	293	612	837	938	957	861	681	298	66
	CDH74	77536	0	1	66	1083	7442	12878	15162	15736	13786	9522	1847	13
	CDH80	48766	0	0	6	104	3290	8558	10698	11272	9466	5113	259	0
Wind	WSAvg	13.9	16.9	15.2	13.8	11.2	12.7	13.8	13.1	12.2	12.9	13.4	16.2	16.1
Precipitation	PrecAvg	7.2	1.4	0.9	1.2	0.7	0.2	0.0	0.0	0.0	0.0	0.2	1.3	1.2
	PrecMax	11.7	4.1	3.2	3.8	2.0	0.8	0.1	0.1	0.1	0.0	1.0	7.2	3.5
	PrecMin	3.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.4	1.0	0.8	0.9	0.6	0.2	0.0	0.0	0.0	0.0	0.2	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.3	70.1	79.2	86.1	101.4	107.5	108.2	106.9	101.0	96.0	85.6	75.5
		MCWB	63.6	62.4	66.1	68.5	81.3	86.1	83.8	83.6	79.7	79.6	72.8	67.5
	2%	DB	68.3	68.7	75.5	82.8	94.8	102.6	103.2	102.7	99.0	94.0	84.3	74.1
		MCWB	62.5	62.1	65.7	69.6	75.8	80.1	79.9	82.0	79.8	78.6	72.5	66.3
	5%	DB	67.6	67.8	73.4	80.5	91.8	99.3	100.5	100.7	97.9	92.4	83.1	72.7
		MCWB	61.8	61.8	64.9	69.5	75.5	78.8	79.9	81.8	79.7	78.2	71.1	64.7
	10%	DB	66.8	66.9	71.9	79.0	89.8	97.1	98.9	99.1	96.8	91.2	81.5	71.3
		MCWB	60.9	61.2	64.3	69.3	75.5	78.4	80.4	82.1	79.7	77.7	69.4	62.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.0	65.0	69.8	74.1	83.8	88.9	89.9	91.4	89.7	85.6	77.4	70.1
		MCDB	68.5	67.9	75.2	81.4	96.4	101.8	95.2	96.7	94.2	92.0	83.2	74.0
	2%	WB	64.5	64.1	67.5	72.7	81.0	86.1	88.8	90.0	87.7	83.5	75.4	68.4
		MCDB	67.2	67.3	72.5	79.3	89.0	95.2	94.5	95.0	94.2	90.4	82.5	72.9
	5%	WB	63.4	63.3	66.6	71.7	79.8	84.4	87.7	89.2	86.4	81.9	73.2	66.8
		MCDB	66.9	66.7	71.2	78.3	87.4	93.7	94.0	94.3	93.4	89.8	81.2	71.7
	10%	WB	62.0	62.4	65.9	70.6	78.5	82.7	86.6	88.5	85.3	80.6	71.3	64.7
		MCDB	66.3	66.1	70.4	77.1	86.3	92.5	93.8	94.1	93.1	88.7	79.8	70.0
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.0	5.8	6.8	8.1	9.6	8.6	7.4	6.3	5.6	5.0	4.9
		MCDBR	4.7	5.3	7.7	9.5	12.0	13.5	11.2	10.2	7.3	6.4	5.9	5.4
		MCWBR	6.0	4.5	4.7	5.6	7.9	7.9	7.6	8.0	8.3	7.5	6.4	6.2
	5% WB	MCDBR	4.4	4.5	5.7	7.9	10.7	11.9	9.0	7.7	6.0	5.8	5.1	4.9
		MCWBR	6.4	4.2	4.5	5.4	7.9	8.8	7.3	6.7	7.6	7.5	6.8	6.7
Clear-Sky Solar Irradiance	taub	0.513	0.560	0.613	0.698	0.694	0.652	0.676	0.626	0.568	0.559	0.544	0.508	
	taud	1.874	1.779	1.664	1.565	1.577	1.632	1.622	1.717	1.819	1.832	1.851	1.898	
	Ebn at Noon	221	223	221	208	210	218	213	222	230	220	210	215	
	Edh at Noon	55	65	77	88	87	82	82	75	65	61	55	51	
All-Sky Solar Radiation	RadAvg	1063	1304	1637	1791	2070	2284	2173	2084	1880	1516	1108	974	
	RadStd	72	88	90	103	89	89	61	48	66	62	87	92	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+0.59	N/A	N/A	+0.97	N/A	N/A	N/A	N/A	+169	+193

Nomenclature: See separate page